



Concomitant optic disk drusen and papilledema due to idiopathic intracranial hypertension in a pediatric cohort

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BACKGROUND	Optic disk drusen (ODD) in pediatric patients typically presents with pseudopapilledema. Diagnosing concomitant papilledema due to idiopathic intracranial hypertension (IIH) in these patients can be challenging. The purpose of this study was to evaluate the incidence and clinical features of papilledema due to IIH among pediatric patients with a new diagnosis of ODD and to discuss the clinical and paraclinical findings that helped diagnose this group.
METHODS	The medical records of children <15 years of age with ODD confirmed by B-scan ultrasound at their first visit over a 4-year period (2019–2022) were reviewed retrospectively. Patients with concurrent IIH were identified, and the demographic and clinical characteristics were reviewed.
RESULTS	A total of 83 children with confirmed ODD at the initial presentation were included, of whom 4 (4.8%) were diagnosed with concomitant IIH. Patients ranged in age from 7 to 15 years; 3 of the 4 were female, and 3 had IIH-related symptoms at presentation (1 was asymptomatic). None of the 4 patients had papilledema greater than Frisen grade 2.
CONCLUSIONS	We recommend that clinicians review pertinent IIH symptoms and risk factors in children with ODD and follow the standard workup for IIH in suspicious cases. In asymptomatic patients with a new diagnosis of ODD, we recommend obtaining a follow-up optic nerve evaluation and optical coherence tomography scan to detect any significant interval change that might serve as a possible indicator of concomitant papilledema. (J AAPOS 2024;28:103806)

Optic disk drusen (ODD) are calcified deposits within the optic nerve head and are found in 0.3%–2% of the pediatric population.^{1–5} ODD tend to stay buried within the optic nerve tissue before the second decade of life, with a higher likelihood of extrusion as patients age into adulthood.⁶ Unlike extruded ODD, buried ODD are difficult to visualize on examination and typically appear as pseudopapilledema, an elevated disk appearance that can resemble papilledema. ODD is often an incidental and asymptomatic finding in children, although it has the potential to cause visual field defects due to calcification bodies compressing nerve fibers and causing subsequent axonal loss later in life.⁶

Idiopathic intracranial hypertension (IIH) or primary pseudotumor cerebri syndrome (PTCS) has a reported incidence of up to 1 in 100,000 to 150,000 in the pediatric population.⁷ Papilledema or swelling of the optic disk secondary to elevated intracranial pressure is a key feature of IIH. Papilledema may cause visual loss from damage to retinal nerve fibers.

Distinguishing between buried ODD and low-grade papilledema due to IIH can be challenging, especially in the pediatric population. While several studies have evaluated the accuracy of different modalities for differentiating between papilledema and ODD, few studies have described the co-occurrence of both conditions in the pediatric group.^{8–12} The reported prevalence of ODD in those diagnosed with idiopathic intracranial hypertension (IIH) ranges from 14.7% to 48%.^{13–15} The purpose of the current study was to investigate the incidence and clinical features of concomitant papilledema due to IIH in a large series of pediatric patients with ODD and to discuss the clinical and paraclinical findings that helped diagnose this group.

Subjects and Methods

We performed a retrospective study of patients who were diagnosed with ODD during the years 2019–2022 at the pediatric neuro-ophthalmology clinic of the Callahan Eye Hospital at the

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Submitted August 29, 2023.

Revision accepted November 6, 2023.

Published online January 10, 2024.

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1091-8531/\$36.00

<https://doi.org/10.1016/j.jaapos.2023.11.009>

Table 1. Demographic and clinical features of patients with optic disk drusen and concomitant idiopathic intracranial hypertension

Case	Age, years	Sex	Symptoms	Opening pressure, cm H ₂ O	Other features
1	15	F	Recent headaches, positional visual changes	28	Improved headache after lumbar puncture
2	7	M	None	30	Decreased RNFL thickness and decreased blurriness of optic disk margins after treatment with acetazolamide
3	12	F	Recent headaches with nausea, dizziness, pulsatile tinnitus	36	
4	12	F	Recent headaches, pulsatile tinnitus	30	

RNFL, retinal nerve fiber layer.

University of Alabama at Birmingham (UAB). This study was approved by the UAB Institutional Review Board and complied with the US Health Insurance Portability and Accountability Act of 1996 and the tenets of the Declaration of Helsinki.

A search engine was used to screen the electronic medical records for patients <15 years of age and ICD-10 codes for “optic disc drusen,” “anomalous optic nerve,” or “pseudopapilledema.” Inclusion criteria included ODD confirmed by B-scan ultrasound at the first

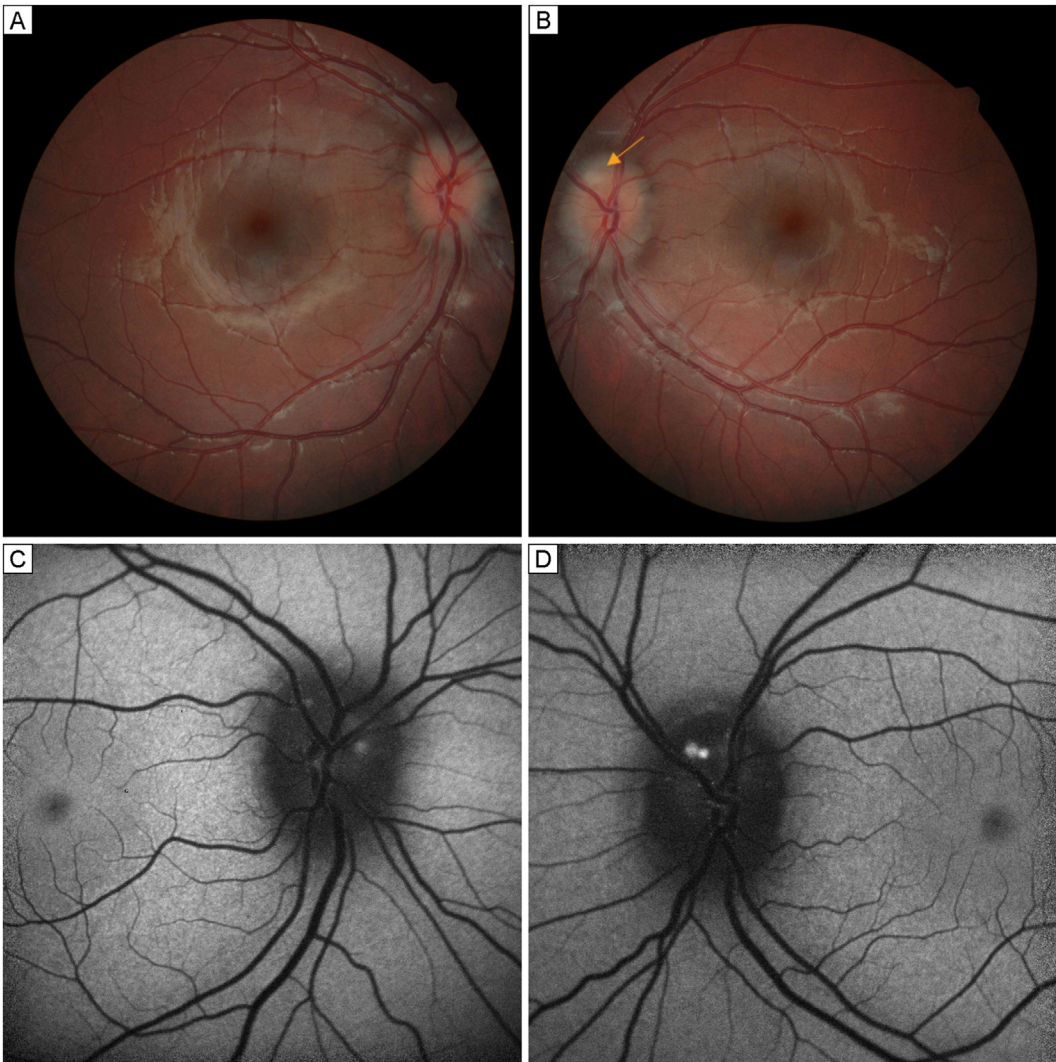


FIG 1. A 12-year-old girl with a body mass index of 20, referred for evaluation of optic nerves. She had a bilateral blurred margin disk (A-B), with apparent drusen that was evident in the superior part of the left disk (B, arrow). Optic disk drusen was confirmed with autofluorescence and B-scan ultrasound (C-D). She reported having a recent new type of headache and occasional pulsatile tinnitus. She was subsequently diagnosed with idiopathic intracranial hypertension.

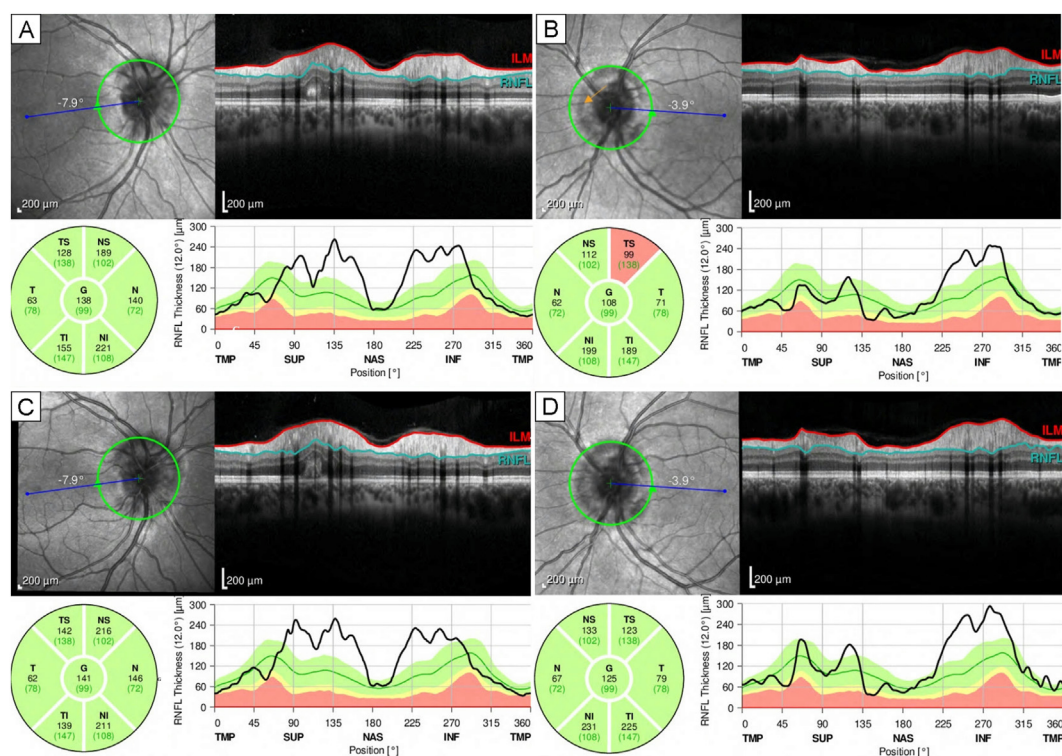


FIG 2. At her 2-month follow-up, the patient whose baseline fundus photographs are shown in [Figure 1](#) reported increasing symptoms, and the retinal nerve fiber layer on OCT showed an interval increase in average thickness from baseline (A-B) to follow-up (C-D). The lumbar puncture revealed an opening pressure of 30 cm H₂O and normal cerebrospinal fluid analysis.

visit. All enrolled subjects had undergone a full ophthalmic examination. This included a dilated fundus examination and indicated ancillary tests, such as fundus photography, autofluorescence, B-scan ultrasound, and optical coherent tomography (OCT; Heidelberg, Carlsbad, CA). The OCT scans included 12° peripapillary scans to evaluate the retinal nerve fiber layer (RNFL). In order to have comparable serial RNFL OCT scans, the operator manually refined the outlines of the retinal nerve fiber layer using the Heidelberg software in the case of apparent automated segmentation errors. Examinations, including OCT, were repeated in all patients within 6 weeks of the initial evaluation. The Frisén grading system was used to classify the papilledema severity.¹⁶

Investigation into the possibility of increased intracranial pressure was pursued if the patients presented with pertinent symptoms (headache, pulsatile tinnitus, positional transient blurred vision), if they had evident concurrent papilledema, or if follow-up examinations revealed increased apparent disk edema (increased elevation, hyperemia, blurred margins, or an interval increase in average RNFL thickness on OCT of $\geq 5 \mu\text{m}$). This investigation included magnetic resonance imaging (MRI) of the orbits and brain with and without gadolinium contrast, magnetic resonance venography of the brain, and lumbar puncture for measurement of opening pressure and complete cerebrospinal fluid analysis. Patients were diagnosed with IIH based on Friedman and colleagues' diagnostic criteria.¹⁷

Results

Of the 83 patients with confirmed ODD at presentation, 4 (4.8%) were diagnosed with concomitant IIH. Age range of these 4 patients (3 females) was 7-15 years ([Table 1](#)). Three had IIH-related symptoms at presentation, while one was asymptomatic. The asymptomatic patient was the youngest, the only male subject, and, in his case concomitant disk edema was suspected due to the changes in the characteristics of the optic disk and increased RNFL thickness in OCT on follow-up examinations. None of the 4 patients had papilledema greater than Frisén grade 2 ([Figures 1-3](#)). All subjects were treated with oral acetazolamide, which reduced RNFL thickness, decreased disk edema, and improved symptoms in all 4 cases. Of the 83 patients with ODD, only these 4 underwent a lumbar puncture for suspicion of IIH.

Discussion

Our study found the prevalence of IIH and papilledema to be 4.8% among children with ODD, which is considerably higher than the general prevalence of pediatric IIH (1 in 100,000 to 150,000).⁷ A few reports describe increased rates of ODD among patients diagnosed with IIH.¹³⁻¹⁵ Genizi

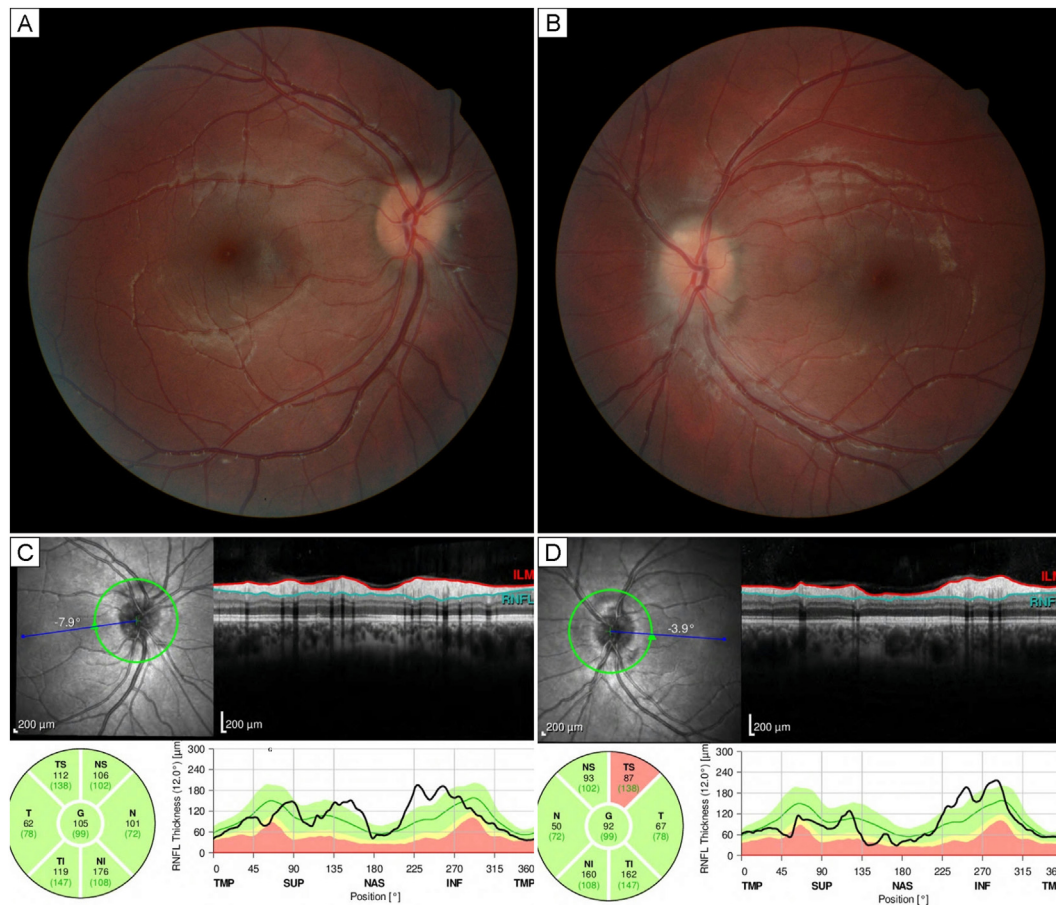


FIG 3. After the patient started acetazolamide 500 mg twice a day, she showed improvement of her symptoms and her papilledema (A-D).

and colleagues¹⁴ reported a 14.7% rate of ODD in their series of pediatric IIH patients, and Gospe and colleagues¹⁵ found a rate of 48% of buried ODD among their pediatric IIH patients using enhanced-depth imaging OCT (EDI-OCT). Gospe and colleagues¹⁵ postulated that eyes with small foramina openings at the lamina cribrosa may have axoplasmic flow stasis, resulting in intramitochondrial calcifications with subsequent extracellular extrusion. Such changes may predispose the optic nerve to be susceptible to the compressive effects of increased intracranial pressure and subsequent papilledema.¹⁵ Our study, conversely, surveyed the rate of IIH among patients with ODD, which has not been previously evaluated, to our knowledge.

In most pediatric patients, ODD is generally asymptomatic and discovered incidentally during routine eye examination. Therefore, symptoms associated with IIH, such as recent headaches, positional symptoms, and pulsatile tinnitus, warrant further investigation for increased intracranial pressure. However, pediatric patients with IIH, especially younger patients, often deny having symptoms. Therefore, we recommend repeat examinations at 1-2 month intervals to detect subtle changes in disk appearance or RNFL on OCT that may be suggestive of true edema in otherwise asymptomatic patients.¹⁸

This study demonstrates that some pediatric patients with newly diagnosed pseudopapilledema due to ODD may concomitantly suffer from IIH with true papilledema. While it is essential to consider ODD as a differential diagnosis of IIH-related papilledema in order to avoid unnecessary neuroimaging and lumbar puncture, we suggest that recognizing the potential coexistence of ODD with true disk edema due to increased intracranial pressure is equally important.

We recommend review of pertinent IIH symptoms and risk factors in children with ODD and suspected pseudopapilledema, with the standard workup for IIH warranted in clinically suspicious cases. In asymptomatic patients with a new diagnosis of ODD, we recommend longitudinal follow-up to confirm stability of the optic nerve appearance; our preference is to re-evaluate the optic disk and obtain a second OCT scan at an interval of 1-2 months, with any significant changes in the appearance of the disk or RNFL thickness serving as a possible indicator of concomitant papilledema. Although the subjects with papilledema in this series were all diagnosed with IIH or primary pseudotumor cerebri syndrome, our findings can be generalized to papilledema due to secondary types of pseudotumor cerebri syndrome.

The primary limitation of this study is that it is a small, single-center retrospective study. Larger, multicenter prospective cohorts are needed to further explore the physiologic changes that link ODD, papilledema, and IIH, as well as studies to evaluate the management strategies for patients with concomitant diagnoses. Furthermore, recent advancements in OCT imaging have been shown to effectively distinguish papilledema from ODD, with potentially higher sensitivity and specificity than B-scan imaging and ophthalmoscopy alone in diagnosing buried and noncalcified drusen among young patients.^{9,10} Given that we did not use the EDI-OCT as the primary tool for diagnosis of ODD, we may have missed potential study subjects with concomitant IIH and ODD. Indeed, the reported prevalence of concomitance may rise with advancements in imaging modalities.

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